

Invisible Fragmented Cryptographic Fingerprints for Protecting Images from Unauthorized AI Training

Principal Investigator

Shiqing Ma

Associate Professor

Manning College of Information & Computer Sciences
University of Massachusetts, Amherst

☎ (413) 545-9266

✉ shiqingma@cs.umass.edu

🌐 people.cs.umass.edu/~shiqingma

[Google Scholar](#)

Student Researcher

Mingzhe Li

Ph.D. Student

Manning College of Information & Computer Sciences
University of Massachusetts, Amherst

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Abstract

Abstract – Generative AI systems have raised urgent security concerns about unauthorized use of protected content for model training. A promising defense approach is to poison the dataset so that models trained on such images cannot converge or generate misaligned images. Existing methods either introduce noticeable artifacts that degrade image quality or are vulnerable to simple image transformations that remove the protection. We propose a novel content protection framework that proactively embeds **fragmented cryptographic fingerprints** into visual data to defend against unauthorized text-to-image model training. Our approach introduces multiple imperceptible fingerprint perturbations per image, each derived from cryptographic sequences, such that any model trained on these images will inherently learn our unique signatures. We design a **robust statistical detection** method to identify models that have ingested the protected data by detecting the presence of multiple embedded fingerprints in the model’s outputs or internal representations. The fingerprints are carefully crafted to be resilient to common attacks like cropping, compression, fine-tuning, or pruning, thereby ensuring reliable tracing even under adversarial conditions, advancing state-of-the-art robustness. We further optimize the fingerprint generation process using neural cryptography techniques to scale to large datasets efficiently.

Introduction & Problem Statement

Problem Statement & Formulation. The proposal aims to address the urgent need for technical solutions to protect content from unauthorized AI training. Formally, we consider a scenario in which a data owner has a set of images $D = \{x_i\}$ intended for public release (e.g., on the web or in a dataset) but wants to prevent or detect their unauthorized use in training a generative model M . An adversary (unauthorized model trainer) may collect these images and include them in training M (e.g., a diffusion model) without consent. Our goal is to empower the data owner with a mechanism to either (a) *deter effective training on the protected dataset D* , or (b) *enable reliable detection* that M has utilized D in its training set. We target the training phase of the unauthorized model as the point of defense, as opposed to inference-time content filters.

To this end, we formulate a protective transformation $\mathcal{T}$ on images such that the transformed dataset $D' = \{\mathcal{T}(x_i)\}$ is (i) nearly indistinguishable from D by humans or automated content scanners (**stealth**), and (ii) contains hidden signals that will be *memorized* by any model trained on D' . We define a detection function $\mathcal{F}$ that, given query access to a trained model M (either white-box access to weights or black-box access to generate images), can statistically infer whether M was trained on D' . $\mathcal{F}$ should yield a high-confidence indication of unauthorized usage even if M 's trainer deploys countermeasures like data augmentations, fine-tuning on additional data, or partial editing to remove signals.

Threat model. The adversary gathers public images (some protected) to train a generative model (e.g., a diffusion model), and may use standard augmentations (cropping, resizing, color jitter), JPEG compression, and limited fine-tuning on extra clean data. We evaluate both black-box (query-only) and white-box (weights / activations) settings. The defender's secret key remains unknown to the attacker, who also lacks large paired sets of original and fingerprinted images (no fingerprint oracle). Robustness is measured versus attacker capability: protected fraction, augmentation strength, and fine-tuning extent.

Limitations of Prior Work. Existing defenses fall into three families, each insufficient for our goals: (i) classical watermarking is often removable (cropping, noise, compression) and not designed to survive model training [8]; (ii) statistical tracing (e.g., radioactive data) [3] shows promise but has been validated mainly on classifiers and may weaken under generative model augmentations; (iii) poisoning / unlearnable examples degrade unauthorized training [1] but do not yield affirmative proof of misuse and can be mitigated by data cleaning or robust optimization. Recent provenance methods for generative settings (anti-neuron color transforms [8], clean-label backdoor watermarking [4]) rely on single global or label-dependent triggers, were evaluated on classifiers, and are vulnerable to fine-tuning or model editing. These gaps motivate a defense that is stealthy, fragmentation-based, model-agnostic, and statistically auditable after training.

Proposed Research

Our approach consists of two synergistic components: (1) **Fragmented Cryptographic Fingerprints** embedded in the training data, and (2) a **Robust Statistical Detection Framework** for identifying models trained on protected data.

Step 1: Fragmented Cryptographic Fingerprints We embed multiple hidden cryptographic fingerprint signals per image (fragmented weak watermarks) rather than a single signature [3, 8]. For image x we generate m perturbations $\delta_1, \dots, \delta_m$ from a master secret key k via independent cryptographically seeded bit streams shaped to match x , optionally constrained for spatial smoothness / frequency profiles favored by DNNs [4]. We form $x' = x + \sum_{i=1}^m \alpha_i \delta_i$, with small α_i controlling perceptual invisibility. Leveraging inherent backdoor observations [5], we align some δ_i with naturally learnable features (subtle textures, high-frequency lattices) to create natural triggers easily absorbed during training. We further search perturbations maximizing memorization while imperceptible via $\max_{\delta_i} \mathbb{E}_{M \sim \text{train}(D')} [g_i(M)]$, where $g_i(M)$ measures the learned imprint (e.g., probe response).

A key technical challenge is generating and embedding these cryptographic fingerprints efficiently at scale. Cryptographic operations (e.g., secure PRNGs, hashes) are typically computationally intensive, and adding them naively to a data preprocessing pipeline could be a bottleneck for large datasets. We will leverage our prior work, *Repurposing Neural Networks for Efficient Cryptographic Computation* [2], to address this. Jin *et al.* (2025) showed that designed GPU-friendly crypto primitives to perform cryptographic computations much faster than standard CPU implementations, by exploiting software and hardware optimizations. More specifically, our proof-of-concept implementations in TensorFlow demonstrate substantial performance improvements: encryption speeds for AES, Chacha20, and Salsa20 show increases of up to $4.09\times$, $5.44\times$, and $5.06\times$, respectively, compared to existing GPU-based cryptographic

solutions written by human experts.

Advantages: (1) Resilience to localized tampering: cropping or partial transformations seldom remove all fragments; eliminating every fingerprint would require destructive distortion that harms training utility. (2) Multi-signal fusion: diverse co-occurring weak signals jointly reinforce learnability; if compression diminishes one, others persist. (3) Avoid single point of failure versus one-pattern schemes.

Security property. The fingerprints are cryptographically unlinkable without the key. Each image receives a unique composite; indices i may map to consistent frequency templates, while the per-image secret phase or orientation varies under k . Consequently, clustering protected images is difficult without many samples or the key. Possession of k enables definitive later verification (see the following sections).

Component 2: Robust Statistical Detection Framework The ultimate step is to design a procedure $\mathcal{F}$ for detecting our fingerprints in a suspect model M . We consider both *white-box* access (we can inspect M 's weights or internal neurons) and *black-box* access (we can query M to generate images or obtain likelihoods). Our detection framework will be formulated as a statistical hypothesis test:

$H_0 : M$ was *not* trained on our protected data, $H_1 : M$ was trained on our protected data.

The test needs to reliably reject H_0 (with significance α) when H_1 is true, even if only a fraction of M 's training data contained our fingerprints.

To exploit our multiple fingerprints, we adopt an approach of aggregating evidence from each fingerprint pattern. Suppose we have m fingerprint patterns, for each pattern i , we can design a *probe* that elicits M 's reaction to that pattern. In a simple white-box scenario, this could be done by computing a correlation between M 's weights/activations and the fingerprint δ_i (similar to checking for a known watermark in a neural weight matrix [3]). In a black-box generative scenario, we could attempt to generate images that emphasize fingerprint i and see if M reproduces aspects of it. One promising technique is to perform *latent optimization*: initialize a latent code for M 's generator and optimize it (via gradient descent) to maximize the presence of fingerprint i in the output image. If M has learned δ_i , it should be easier to generate images containing δ_i (with low loss) compared to a model that never saw δ_i .

Once we obtain a metric or score s_i for each fingerprint i on model M , we combine them into a test statistic $T(M) = f(s_1, \dots, s_m)$. A simple choice could be $T = \frac{1}{m} \sum_{i=1}^m s_i$ (the average response to fingerprints) or the maximum response $\max_i s_i$. We will explore more sophisticated combinations, possibly weighting fingerprints by their expected strength (some fingerprints might be more salient than others). Under H_0 , we expect M to have only random or baseline responses to our fingerprints. Under H_1 , M should show significantly higher responses. We will empirically derive the null distribution of T by testing on models known not to have seen our data (or by theoretical modeling if feasible), and then choose a threshold τ such that $\Pr_{H_0}(T > \tau) < \alpha$. Concretely, for each fingerprint i we compute a normalized response score s_i . Our default test statistic is the mean response $T(M) = \frac{1}{m} \sum_{i=1}^m s_i$, and we reject the null hypothesis if $T(M) > \tau$, where τ is set from the empirical null (e.g., the $1-\alpha$ quantile estimated on a collection of models trained without exposure to protected images). If the suspect model yields $T(M) > \tau$, we declare a detection of unauthorized data usage.

Our approach is inherently robust to partial trigger removal attacks. Even if the adversary manages to suppress some fingerprint signals in M , as long as at least one or a combination of a few fingerprints remain, the statistic T is likely to be above the expected range of an untouched model. By analogy, Tang *et al.* used non-parametric hypothesis testing to assert the presence of a backdoor function by comparing model outputs with and without triggers [4]. We aim to extend such tests to a multi-trigger scenario. For example, we could perform a Wilcoxon signed-rank test [4] on pairs $(s_i^{\text{trigger}}, s_i^{\text{base}})$ for each i , where s_i^{trigger} is the model's output probability (or some performance measure) when fingerprint i is activated, and s_i^{base} is in a normal condition. A significant shift in these scores across fingerprints would indicate H_1 . Because we have multiple independent fingerprints, our detection can tolerate a few s_i being low (if those particular patterns were pruned out) as long as the majority show a detectable effect.

We evaluate detection in both white- and black-box regimes. With weight access we directly correlate filters or activations with fingerprints [8]. Under black-box (API) access we issue adaptive queries, optimizing outputs to elicit fingerprint responses, leveraging prior generative origin attribution methods [6, 7]. The result is a statistical yes/no decision on unauthorized use to support enforcement actions.

Intellectual Merit: We introduce *fragmented cryptographic fingerprints*, distributing multiple cryptographically derived perturbations across each image to eliminate single points of failure; we develop a rigorous statistical detection framework casting attribution as a formal hypothesis test with controlled error rates and combining multi-fingerprint evidence via trimmed means, Fisher's method, and empirical Bayes; we advance cryptographic innovation by leveraging neural cryptography (e.g., CryptoNN speedups) to make large-scale fingerprint generation feasible.

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